

KNIGHT.VAULT — USER MANUAL & RUNTIME SPECIFICATION

A Comprehensive Handbook for Operating, Deploying, and Troubleshooting Confidential Zero-Knowledge Settlement Vaults on Midnight Network

Target Network: Midnight Preprod	DApp Name: KNIGHT.VAULT
Smart Contract: Compact Circuits (payment.compact)	ZK Proof Engine: Docker Local Port :6300
Wallet API: 1AM DApp Connector	Architect: Kshitij Adakane

1. Developer Setup & Prerequisites

To execute transactions and run proof synthesis locally, the developer environment requires three primary subsystems:

- **WSL2 (Ubuntu 22.04 LTS):** Required on Windows to host the native Linux Compact toolchain and prover libraries.
- **Docker ProofServer (Port 6300):** Synthesizes zero-knowledge proofs on your machine in 2-4 seconds.
- **1AM Browser Wallet:** Manages unshielded/shielded keys, signs transactions, and relays sponsored dust fees.

1.1 Docker ProofServer Deployment & Port Verification

Run the proof server container and ensure it is listening on localhost:

```
docker run -d --name proof-server -p 6300:6300 midnightnetwork/proof-server:latest
# Health check command:
curl http://localhost:6300/health
```

WSL2 Memory Note: Ensure your .wslconfig has at least 8GB-12GB RAM allocated. Proving large ZK circuits may fail with an unhandled exit code if memory is constrained below 4GB.

2. User Operations: Deposit & Withdraw

2.1 Depositing Funds (Zero-Gas Payer Flow)

1. Open the KNIGHT.VAULT dashboard at <http://localhost:5173/>.
2. Click '**Connect 1AM Wallet**' and authorize permissions on the popup modal.
3. Enter the deposit amount (e.g. 50 tNIGHT) in the '**Inflow Channel**' card.
4. Click '**Execute Confidential Inflow**'. The 1AM wallet balances the transaction with sponsored dust automatically.

2.2 Zero-Knowledge Withdrawal Flow

1. Navigate to the '**Private Liquidity Release**' card.
2. Enter the withdrawal amount and recipient unshielded address (or public key).

3. Click '**Authorize ZK Withdrawal**'. The local ProofServer will compute the witness proof.
4. Once synthesized, the transaction is committed to Midnight Preprod with zero private data leaked.

3. Real-World Troubleshooting & FAQs

Issue / Error	Root Cause & Recommended Solution
ProofServer Unreachable (Port 6300)	Docker container is stopped or port is blocked by Windows Firewall. Run docker restart proof-server and verify with curl http://localhost:6300/health.
1AM Wallet 'offset: null' Error	Known bug in Midnight Indexer v4 GraphQL schema when pagination offset is null. The KNIGHT.VAULT client includes an automatic filter polyfill.
Insufficient Dust Balance	The 1AM wallet needs a small unshielded balance to derive dust. Request 10 tNIGHT from the Midnight faucet to refresh coin notes.
Vite / Browser Polyfill Errors	Node.js stream/crypto modules must be polyfilled in browser builds. Ensure vite-plugin-node-polyfills is included in vite.config.ts.

4. Summary & Architect Credits

KNIGHT.VAULT demonstrates modern, institutional-grade zero-knowledge engineering. Designed and built by Kshitij Adakane.

Open Source Codebase: <https://github.com/KSHITIJadakane/KNIGHT.VAULT>